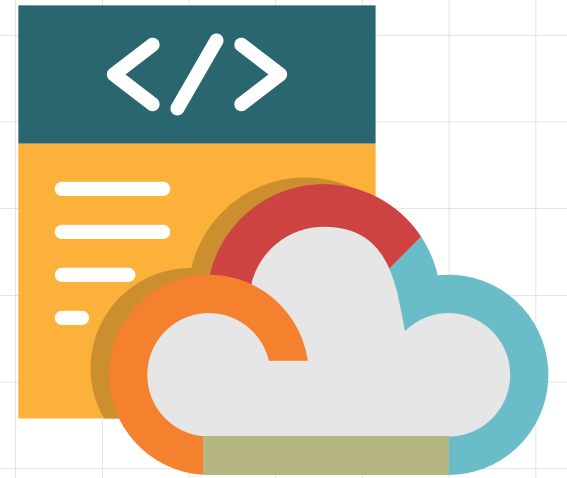


Top GCP Services Every Data Engineer Must Know

(With Real-World Use Cases)



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01 BIGQUERY

Type: Serverless Data Warehouse

Use Case:

Massive-scale data analysis using SQL.

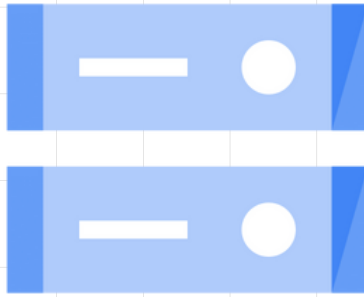
Real-Life Example:

Spotify uses BigQuery to analyze user listening data for recommendation engines and real-time music trends.



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02 CLOUD STORAGE

Type: Object Storage (Data Lake Layer)

Use Case:

Store raw, structured, semi-structured, or unstructured data (CSV, JSON, Parquet, etc.).

Real-Life Example:

AirAsia stores clickstream and flight log data in Cloud Storage before processing it in Dataflow.



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03 CLOUD DATA FUSION

Type: Fully Managed, Code-Free ETL/ELT Tool

Use Case:

Design, build, and manage data pipelines with a visual interface.

Real-Life Example:

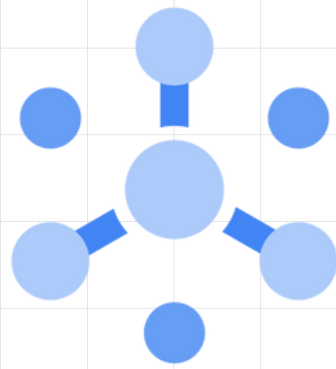
Vodafone uses Cloud Data Fusion to build centralized data pipelines from multiple on-premise sources and cloud applications into BigQuery, enabling real-time analytics and customer insights across their telecom services.



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04 PUB/SUB

Type: Messaging/Event System

Use Case:

Decouple microservices and stream event data.

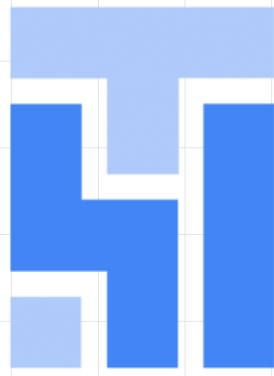
Real-Life Example:

Snapchat uses Pub/Sub to stream events like user snaps, messages, and interactions for analytics and personalization.



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05 CLOUD COMPOSER

Type: Workflow Orchestration
(Airflow)

Use Case:

Schedule and orchestrate data pipelines.

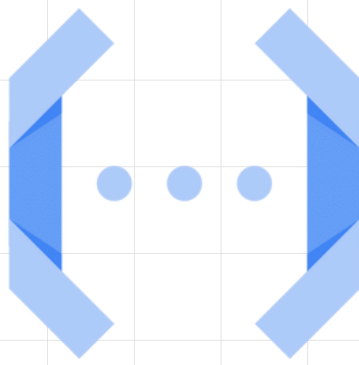
Real-Life Example:

Philips Healthcare uses Cloud Composer to schedule ingestion of medical IoT data into BigQuery for analysis.



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06 CLOUD FUNCTIONS

Type: Serverless Functions

Use Case:

Trigger small units of logic based on events (e.g., file uploads).

Real-Life Example:

The New York Times triggers Cloud Functions to auto-tag articles when editors upload new content to Cloud Storage.



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07 CLOUD SQL

Type: Managed Relational Databases

Use Case:

Store operational, transactional data for reporting and analytics.

Real-Life Example:

Zebra Medical Vision uses Cloud SQL to store medical scan metadata, which is later analyzed through BigQuery pipelines.



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08

BIGTABLE

Type: NoSQL Wide-Column Store

Use Case:

Store high-velocity, time-series data.

Real-Life Example:

Google Ads uses Bigtable internally to manage billions of rows of time-series ad performance data in near-real-time.



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09

DATAPROC

Type: Managed Spark/Hadoop Cluster

Use Case:

Legacy Hadoop/Spark workflows for ETL or ML tasks.

Real-Life Example:

Pandora used Dataproc to migrate legacy Hadoop jobs into GCP while improving cost and performance efficiency.



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10

IAM (IDENTITY & ACCESS MANAGEMENT)

Type: Security and Access Control

Use Case:

Role-based access to data and services.

Real-Life Example:

Verily Life Sciences (Alphabet subsidiary) uses IAM to ensure only data engineers can access sensitive healthcare data in BigQuery.



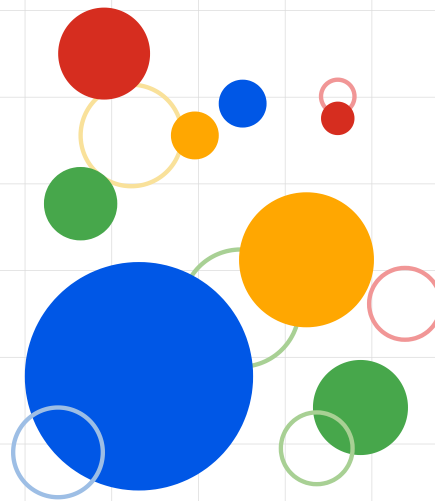
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